

LAB 4:

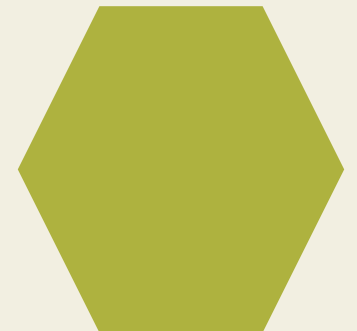
Problem Framing & Modelling I

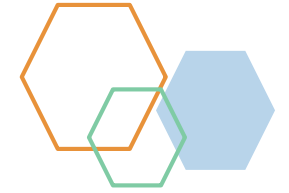
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Lab Instructor

Introduction to Engineering

Fall 2026





Lab Agenda



Recap

Evidence, problem statements, box language, variables, and performance measures from Lecture 4.

15–20 min



Scenario Cards Handed Out

Each team gets a different problem scenario to work through.

5 min



Group Work — Applied Problem Framing

Fill the three-part template: evidence & framing, process & variables, stakeholders & targets.

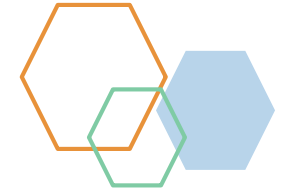
35 min • evaluated



Model Answer Walkthrough

Worked solution for the EV Charging & Battery Test Station scenario.

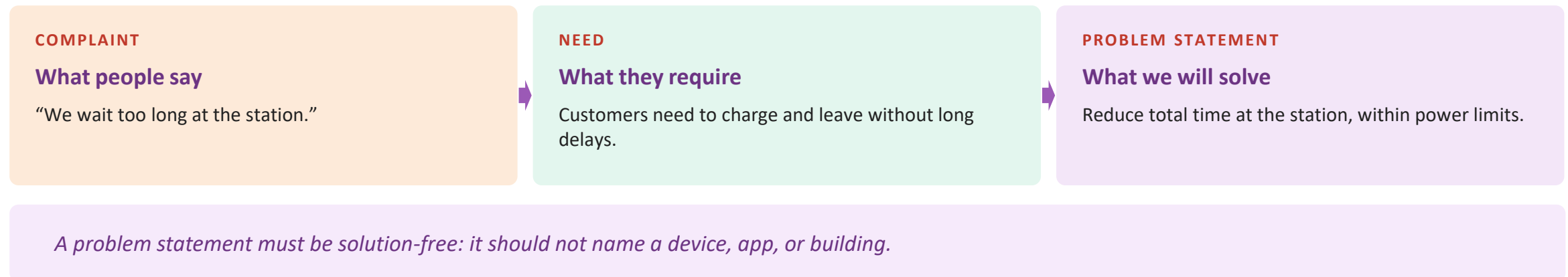
15 min



Evidence Before the Problem

Kind	What it is	Example
Fact	Something observed or measured.	Vehicles arrive at 12 cars/hour (counted).
Inference	A conclusion drawn from facts.	Queues form when all 4 chargers are busy.
Speculation	A guess without evidence yet.	The station may be slower due to safety limits.
Opinion	A personal view or preference.	Customers want shorter waiting times.

From Complaint to Problem Statement



QUICK RECAP

Box Language & Variables

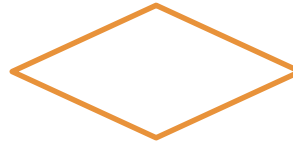
Process flow building blocks



Start / End



Step



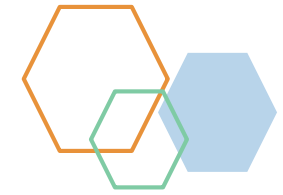
Decision



Flow



Wait / queue



Three kinds of variables

PERFORMANCE

What we judge

Average waiting time, vehicles served per hour, peak power draw.

DESIGN

What we can change

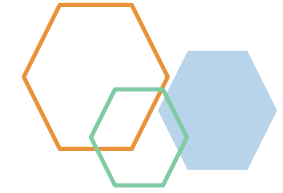
Number of chargers, staffing, diagnostic test duration.

CONTEXT

What we cannot change

Opening hours, grid capacity, safety regulations.

A useful variable answers “how much?” or “how many?” and carries a unit. Every step and arrow in your process diagram suggests something to count, time, or weigh.



Stakeholders, Measures & Targets

Stakeholder	Their idea of success	Candidate variable	Unit
Rider	Short, predictable waits	Waiting time at the stop	minutes
Driver / dispatcher	A schedule that can be kept	Trips completed on time	% of trips
Administration	Good use of the budget	Cost per passenger	currency/rider

DEMAND (D)**Must be met**

Without it the design is not acceptable.

WISH (W)**Nice to have**

Desirable, but not crucial to success.

COURSE TEMPLATE

The performance of **[system]** is judged by **[variable 1]**, **[variable 2]**, **[variable 3]**, each with units and a target, as seen by **[stakeholders]**.

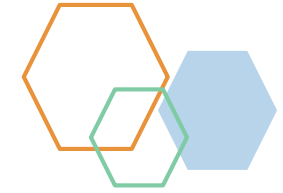
Use numbers wherever possible, even rough ones — “average wait under 10 minutes,” not “short waits.”

GROUP WORK • EVALUATED • 35 MINUTES



Applied Problem Framing

Turn your scenario card into evidence, a process, and measurable targets.



Your Task: Three Parts, One Scenario

Each team receives a different scenario card. Fill the template below for all three parts and be ready to present.



PART A · 10 min

Classify & Frame

Sort evidence into Fact / Inference / Speculation / Opinion, then write a Complaint → Need → Problem Statement.



PART B · 15 min

Map the Process

Draw your process in up to 6–7 steps using box language. Name one variable with its unit per step.



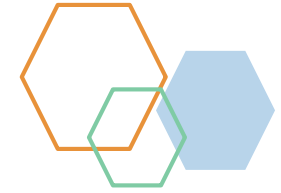
PART C · 10 min

Stakeholders & Targets

Build a stakeholder → success → variable → unit table (3+ rows), then propose must-have and nice-to-have targets.

Scenario cards are printed and handed out — no two teams get the same one.

Graded live: I'll circulate with a checklist and score each team on all three parts before time is up.



Classify & Frame Your Scenario

1 · Sort four statements from your card

Kind	Write one statement from your scenario	Why it belongs here
Fact	...	...
Inference	...	...
Speculation	...	...
Opinion	...	...

2 · Write your problem statement

COMPLAINT

What does the scenario say people complain about?

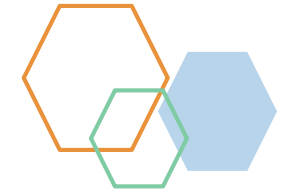
NEED

What do they actually require (not a solution)?

PROBLEM STATEMENT

[Stakeholder] needs [outcome] in [system], judged by [measure], within [constraints].

Solution-free test: does your problem statement name a device, app, vehicle, or building? If yes, rewrite it.



Map the Process & Its Variables

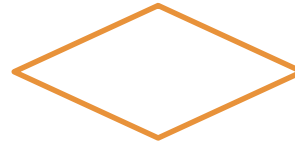
Draw your scenario's process in at most 6–7 steps using the box language below. Under each step, name one variable with its unit.



Start / End



Step



Decision

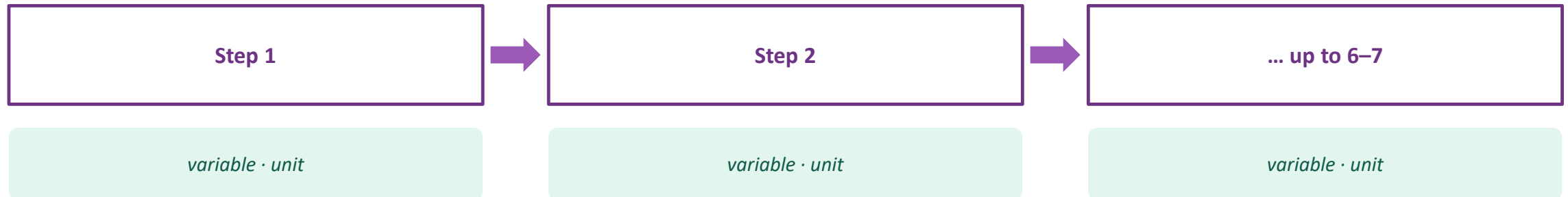


Flow

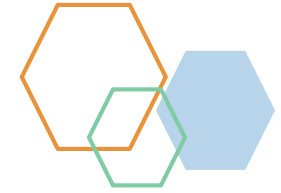


Wait / queue

Format to follow (fill with your own steps)



Use a decision diamond wherever the scenario branches (e.g. a threshold or inspection check), and a dashed box anywhere something waits in a queue.



Stakeholders, Variables & Targets

1 • Map at least 3 stakeholders

Stakeholder	Their idea of success	Candidate variable	Unit
...	...	...	...
...	...	...	...
...	...	...	...

2 • Propose targets for your variables

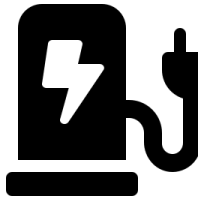
MUST-HAVE (DEMAND)

A target the design cannot fail to meet. Give a number and unit.

NICE-TO-HAVE (WISH)

A target that would be good, but is not required for success.

Write targets as numbers, even rough ones: “peak power draw ≤ 150 kW,” not “low power use.” Each must-have needs a unit.



EV Charging & Battery Test Station

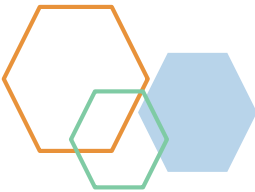
A worked solution applying all three parts of the template.

1

EV Charging and Battery Test Station



An EV charging and battery test station receives electric vehicles for fast charging and battery health checks. Vehicles arrive at an average rate of 12 cars per hour and wait in a queue if all 4 chargers are busy. Each charge takes about 30–40 minutes, followed by a 10-minute battery diagnostic test. If the battery health is below 80%, a technician inspects it. Customers want shorter waiting times, operators are concerned about electricity demand peaks, and the local grid operator worries about high power draw. Some drivers think the station is slower than competitors, possibly due to conservative safety limits.



Classify & Frame

12 cars/hour arrive · 4 chargers · 30–40 min charge + 10-min diagnostic · inspection if battery health < 80%.

Kind	Statement
Fact	Vehicles arrive at an average of 12 cars/hour; the station has 4 chargers.
Inference	Queues form when arrivals outpace the combined charge + diagnostic time of the 4 bays.
Speculation	The station may be slower than competitors, possibly due to conservative safety limits.
Opinion	Customers want shorter waiting times; the grid operator worries about peak power draw.

Complaint → Need → Problem Statement

COMPLAINT

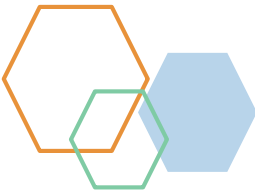
“Drivers think the station is slower than competitors.”

NEED

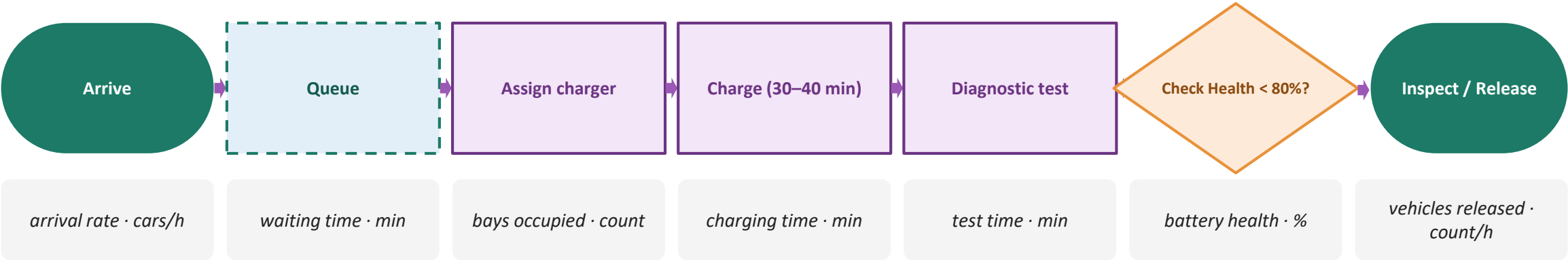
Customers need to charge and verify battery health without a long wait.

PROBLEM STATEMENT

Customers need short, predictable total time at the station, judged by time from arrival to release, within the grid operator's peak-power limit.

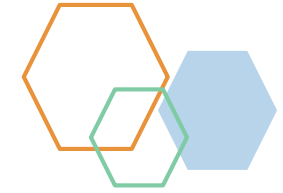


Process Map & Variables



7 steps — within the 6–7 step limit. The decision diamond branches to a technician inspection when battery health falls below 80%.

Step	Variable	Unit / Note
Charge	Power draw per charger	kW — drives the grid-operator's concern
Diagnostic test	Batteries flagged below threshold	% of vehicles tested
Whole process	Total time at station	minutes — the customer's key measure



Stakeholders & Targets

Stakeholder	Their idea of success	Candidate variable	Unit
Customer	Short, predictable total visit	Total time at station	minutes
Station operator	High throughput, managed demand	Vehicles served per hour	count/hour
Grid operator	Low peak power draw	Peak power draw	kW
Technician	Unsafe batteries caught early	Batteries flagged for inspection	% of tests

Must-Have and Nice-to-Have Targets

MUST-HAVE (DEMAND)

- Average total time at station ≤ 45 min at peak hours
- Peak power draw stays within the grid operator's contracted limit (≤ 150 kW)

NICE-TO-HAVE (WISH)

- Average total time at station ≤ 30 min — faster than competitors
- Real-time queue-length display for arriving customers

This is a model answer — your team's scenario is different, and a different (well-justified) set of variables and targets can score just as well.

THANK YOU

Questions?

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